



MATH155: Trigonometry Basics

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Learning Objectives

- Convert between degrees and radians
- Find exact values of sine, cosine, and tangent at standard angles
- Apply the unit circle to evaluate trigonometric functions
- Use SOH-CAH-TOA to solve right triangles

Simplify each expression completely. Show all steps and circle your final answer.

Radian and degree conversion

1. Convert 180° to radians. Express your answer as a fraction times pi.

$$180^\circ$$

Answer: _____

2. Convert the radian measure $1\pi/3$ to degrees.

$$\frac{1\pi}{3}$$

Answer: _____

3. Convert 90° to radians. Express your answer as a fraction times pi.

$$90^\circ$$

Answer: _____

4. Convert the radian measure $3\pi/6$ to degrees.

$$\frac{3\pi}{6}$$

Answer: _____

5. Convert 240° to radians. Express your answer as a fraction times pi.

$$240^\circ$$

Answer: _____



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6. Convert the radian measure $2\pi/4$ to degrees.

$$\frac{2\pi}{4}$$

Answer: _____

7. Convert 180° to radians. Express your answer as a fraction times π .

$$180^\circ$$

Answer: _____

8. Convert the radian measure $1\pi/4$ to degrees.

$$\frac{1\pi}{4}$$

Answer: _____

9. Convert 225° to radians. Express your answer as a fraction times π .

$$225^\circ$$

Answer: _____

10. Convert the radian measure $2\pi/6$ to degrees.

$$\frac{2\pi}{6}$$

Answer: _____

11. Convert 225° to radians. Express your answer as a fraction times π .

$$225^\circ$$

Answer: _____

12. Convert the radian measure $4\pi/2$ to degrees.

$$\frac{4\pi}{2}$$

Answer: _____

13. Convert 60° to radians. Express your answer as a fraction times π .

$$60^\circ$$

Answer: _____



14. Convert the radian measure $1\pi/6$ to degrees.

$$\frac{1\pi}{6}$$

Answer: _____

15. Convert 60° to radians. Express your answer as a fraction times pi.

$$60^\circ$$

Answer: _____

16. Convert the radian measure $3\pi/6$ to degrees.

$$\frac{3\pi}{6}$$

Answer: _____

Unit circle trig values

17. Find the exact value of $\sin(135^\circ)$ using the unit circle.

$$\sin(135^\circ)$$

Answer: _____

18. Find the exact value of $\tan(45^\circ)$ using the unit circle.

$$\tan(45^\circ)$$

Answer: _____

19. Find the exact value of $\sin(120^\circ)$ using the unit circle.

$$\sin(120^\circ)$$

Answer: _____

20. Find the exact value of $\tan(0^\circ)$ using the unit circle.

$$\tan(0^\circ)$$

Answer: _____

21. Find the exact value of $\sin(90^\circ)$ using the unit circle.

$$\sin(90^\circ)$$

Answer: _____



22. Find the exact value of $\tan(120^\circ)$ using the unit circle.

$$\tan(120^\circ)$$

Answer: _____

23. Find the exact value of $\cos(0^\circ)$ using the unit circle.

$$\cos(0^\circ)$$

Answer: _____

24. Find the exact value of $\tan(0^\circ)$ using the unit circle.

$$\tan(0^\circ)$$

Answer: _____

25. Find the exact value of $\cos(90^\circ)$ using the unit circle.

$$\cos(90^\circ)$$

Answer: _____

26. Find the exact value of $\tan(60^\circ)$ using the unit circle.

$$\tan(60^\circ)$$

Answer: _____

27. Find the exact value of $\sin(150^\circ)$ using the unit circle.

$$\sin(150^\circ)$$

Answer: _____

28. Find the exact value of $\tan(60^\circ)$ using the unit circle.

$$\tan(60^\circ)$$

Answer: _____

29. Find the exact value of $\sin(150^\circ)$ using the unit circle.

$$\sin(150^\circ)$$

Answer: _____

30. Find the exact value of $\tan(150^\circ)$ using the unit circle.

$$\tan(150^\circ)$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Radian and degree conversion, Unit circle trig values. All answers verified by independent computation.

Solutions



Radian and degree conversion

1. Convert 180° to radians. Express your answer as a fraction times pi.

$$180^\circ$$

→ Multiply by $\pi/180$: $180^\circ * (\pi / 180) = 6 * \pi / 6$.

→ Simplify the fraction: $1 * \pi / 1$.

Answer: 1π

2. Convert the radian measure $1\pi/3$ to degrees.

$$\frac{1\pi}{3}$$

→ Multiply by $180/\pi$: $(1\pi/3) * (180/\pi) = 1 * 180 / 3$.

→ $= 180 / 3 = 60^\circ$.

Answer: 60°

3. Convert 90° to radians. Express your answer as a fraction times pi.

$$90^\circ$$

→ Multiply by $\pi/180$: $90^\circ * (\pi / 180) = 1 * \pi / 2$.

→ Simplify the fraction: $1 * \pi / 2$.

Answer: $\frac{1\pi}{2}$

4. Convert the radian measure $3\pi/6$ to degrees.

$$\frac{3\pi}{6}$$

→ Multiply by $180/\pi$: $(3\pi/6) * (180/\pi) = 3 * 180 / 6$.

→ $= 540 / 6 = 90^\circ$.

Answer: 90°

5. Convert 240° to radians. Express your answer as a fraction times pi.

$$240^\circ$$

→ Multiply by $\pi/180$: $240^\circ * (\pi / 180) = 4 * \pi / 3$.

→ Simplify the fraction: $4 * \pi / 3$.

Answer: $\frac{4\pi}{3}$

6. Convert the radian measure $2\pi/4$ to degrees.

$$\frac{2\pi}{4}$$

→ Multiply by $180/\pi$: $(2\pi/4) * (180/\pi) = 2 * 180 / 4$.

→ $= 360 / 4 = 90^\circ$.

Answer: 90°



7. Convert 180° to radians. Express your answer as a fraction times π .

$$180^\circ$$

→ Multiply by $\pi/180$: $180^\circ * (\pi / 180) = 2 * \pi / 2$.

→ Simplify the fraction: $1 * \pi / 1$.

Answer: 1π

8. Convert the radian measure $1 * \pi / 4$ to degrees.

$$\frac{1\pi}{4}$$

→ Multiply by $180/\pi$: $(1\pi/4) * (180/\pi) = 1 * 180 / 4$.

→ $= 180 / 4 = 45^\circ$.

Answer: 45°

9. Convert 225° to radians. Express your answer as a fraction times π .

$$225^\circ$$

→ Multiply by $\pi/180$: $225^\circ * (\pi / 180) = 5 * \pi / 4$.

→ Simplify the fraction: $5 * \pi / 4$.

Answer: $\frac{5\pi}{4}$

10. Convert the radian measure $2 * \pi / 6$ to degrees.

$$\frac{2\pi}{6}$$

→ Multiply by $180/\pi$: $(2\pi/6) * (180/\pi) = 2 * 180 / 6$.

→ $= 360 / 6 = 60^\circ$.

Answer: 60°

11. Convert 225° to radians. Express your answer as a fraction times π .

$$225^\circ$$

→ Multiply by $\pi/180$: $225^\circ * (\pi / 180) = 5 * \pi / 4$.

→ Simplify the fraction: $5 * \pi / 4$.

Answer: $\frac{5\pi}{4}$

12. Convert the radian measure $4 * \pi / 2$ to degrees.

$$\frac{4\pi}{2}$$

→ Multiply by $180/\pi$: $(4\pi/2) * (180/\pi) = 4 * 180 / 2$.

→ $= 720 / 2 = 360^\circ$.

Answer: 360°



13. Convert 60° to radians. Express your answer as a fraction times pi.

$$60^\circ$$

→ Multiply by $\pi/180$: $60^\circ * (\pi / 180) = 1 * \pi / 3$.

→ Simplify the fraction: $1 * \pi / 3$.

Answer: $\frac{1\pi}{3}$

14. Convert the radian measure $1 * \pi / 6$ to degrees.

$$\frac{1\pi}{6}$$

→ Multiply by $180/\pi$: $(1\pi/6) * (180/\pi) = 1 * 180 / 6$.

→ $= 180 / 6 = 30^\circ$.

Answer: 30°

15. Convert 60° to radians. Express your answer as a fraction times pi.

$$60^\circ$$

→ Multiply by $\pi/180$: $60^\circ * (\pi / 180) = 1 * \pi / 3$.

→ Simplify the fraction: $1 * \pi / 3$.

Answer: $\frac{1\pi}{3}$

16. Convert the radian measure $3 * \pi / 6$ to degrees.

$$\frac{3\pi}{6}$$

→ Multiply by $180/\pi$: $(3\pi/6) * (180/\pi) = 3 * 180 / 6$.

→ $= 540 / 6 = 90^\circ$.

Answer: 90°



Unit circle trig values

17. Find the exact value of $\sin(135^\circ)$ using the unit circle.

$$\sin(135^\circ)$$

→ Locate 135° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (-\sqrt{2}/2, \sqrt{2}/2)$.

→ $\sin(135^\circ) = \sqrt{2}/2$.

Answer: $\sin(135^\circ) = \sqrt{2}/2$

18. Find the exact value of $\tan(45^\circ)$ using the unit circle.

$$\tan(45^\circ)$$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 45° : $\sin = \sqrt{2}/2$, $\cos = \sqrt{2}/2$.

→ $\tan(45^\circ) = \sqrt{2}/2 / \sqrt{2}/2 = 1$.

Answer: $\tan(45^\circ) = 1$

19. Find the exact value of $\sin(120^\circ)$ using the unit circle.

$$\sin(120^\circ)$$

→ Locate 120° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (-1/2, \sqrt{3}/2)$.

→ $\sin(120^\circ) = \sqrt{3}/2$.

Answer: $\sin(120^\circ) = \sqrt{3}/2$

20. Find the exact value of $\tan(0^\circ)$ using the unit circle.

$$\tan(0^\circ)$$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 0° : $\sin = 0$, $\cos = 1$.

→ $\tan(0^\circ) = 0 / 1 = 0$.

Answer: $\tan(0^\circ) = 0$

21. Find the exact value of $\sin(90^\circ)$ using the unit circle.

$$\sin(90^\circ)$$

→ Locate 90° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (0, 1)$.

→ $\sin(90^\circ) = 1$.

Answer: $\sin(90^\circ) = 1$

22. Find the exact value of $\tan(120^\circ)$ using the unit circle.

$$\tan(120^\circ)$$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 120° : $\sin = \sqrt{3}/2$, $\cos = -1/2$.

→ $\tan(120^\circ) = \sqrt{3}/2 / -1/2 = -\sqrt{3}$.

Answer: $\tan(120^\circ) = -\sqrt{3}$



23. Find the exact value of $\cos(0^\circ)$ using the unit circle.

$$\cos(0^\circ)$$

→ Locate 0° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (1, 0)$.

→ $\cos(0^\circ) = 1$.

Answer: $\cos(0^\circ) = 1$

24. Find the exact value of $\tan(0^\circ)$ using the unit circle.

$$\tan(0^\circ)$$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 0° : $\sin = 0$, $\cos = 1$.

→ $\tan(0^\circ) = 0 / 1 = 0$.

Answer: $\tan(0^\circ) = 0$

25. Find the exact value of $\cos(90^\circ)$ using the unit circle.

$$\cos(90^\circ)$$

→ Locate 90° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (0, 1)$.

→ $\cos(90^\circ) = 0$.

Answer: $\cos(90^\circ) = 0$

26. Find the exact value of $\tan(60^\circ)$ using the unit circle.

$$\tan(60^\circ)$$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 60° : $\sin = \sqrt{3}/2$, $\cos = 1/2$.

→ $\tan(60^\circ) = \sqrt{3}/2 / 1/2 = \sqrt{3}$.

Answer: $\tan(60^\circ) = \sqrt{3}$

27. Find the exact value of $\sin(150^\circ)$ using the unit circle.

$$\sin(150^\circ)$$

→ Locate 150° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (-\sqrt{3}/2, 1/2)$.

→ $\sin(150^\circ) = 1/2$.

Answer: $\sin(150^\circ) = 1/2$

28. Find the exact value of $\tan(60^\circ)$ using the unit circle.

$$\tan(60^\circ)$$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 60° : $\sin = \sqrt{3}/2$, $\cos = 1/2$.

→ $\tan(60^\circ) = \sqrt{3}/2 / 1/2 = \sqrt{3}$.

Answer: $\tan(60^\circ) = \sqrt{3}$



29. Find the exact value of $\sin(150^\circ)$ using the unit circle.

$\sin(150^\circ)$

→ Locate 150° on the unit circle.

→ The coordinates at this angle are $(\cos, \sin) = (-\sqrt{3}/2, 1/2)$.

→ $\sin(150^\circ) = 1/2$.

Answer: $\sin(150^\circ) = 1/2$

30. Find the exact value of $\tan(150^\circ)$ using the unit circle.

$\tan(150^\circ)$

→ Use the identity $\tan(\theta) = \sin(\theta) / \cos(\theta)$.

→ At 150° : $\sin = 1/2$, $\cos = -\sqrt{3}/2$.

→ $\tan(150^\circ) = 1/2 / -\sqrt{3}/2 = -\sqrt{3}/3$.

Answer: $\tan(150^\circ) = -\sqrt{3}/3$

